
IoT Design, Data, and Applications
Technical Design Constraints
Power Constraints
Battery-powered remote deployment
High energy use: Wireless communication
Low-power microcontrollers: ARM Cortex-M, ATmega328
Strategies: Sleep modes and energy harvesting
Processing and Memory Constraints
RAM limits: A few KBs
Clock speed: Tens of MHz
Strategies: Edge computing and cloud analytics
Communication Constraints
Bluetooth/BLE range: 10–50 m
Wi-Fi range: 50–100 m
ZigBee range: 10–100 m (mesh capable)
LoRa range: Up to 15 km
Trade-off: Long range equals lower data rate and higher latency
Network Topology Constraints
Star topology: Single gateway used in Wi-Fi
Mesh topology: Devices relay via neighbors in ZigBee
Tree topology: Hierarchical connection in industrial IoT
Environmental and Physical Constraints
Uncontrolled environments: Temperature, humidity, dust, vibration
Measures: IP-rated enclosures IP65 and IP68
Operating range: -40°C to +85°C
Cost Constraints
Affordable SoCs: ESP8266 for basic sensing
Modular designs for hardware reuse
Security Constraints
Challenges: Resource-constrained encryption limits
Strategies: Lightweight encryption AES-128 and ECC
Reliability and Maintenance
Self-diagnostics and remote firmware updates (OTA)
IoT Networks and Layers
Network Layers
Perception Layer: Physical devices, sensors, and actuators
Network Layer: Transmission technologies like Wi-Fi and LoRa
Application Layer: Cloud services, dashboards, and user apps
Network Design Constraints
Bandwidth: Data compression and batching
Latency: Edge processing and QoS protocols
Interference: Frequency hopping and mesh routing
Security: Encrypted channels: TLS and DTLS
Scalability: Hierarchical or mesh networks
Reliability: Redundant paths and retransmission
Data Representation and Visualization
Data Representation
Goals: Reduce size and maintain interoperability
Text Formats: JSON and XML
Binary Formats: CBOR, MessagePack, and Protocol Buffers
Semantic Representation: RDF and OWL for machine understanding
Storage Options: Local, edge, and cloud databases
Data Visualization
Purpose: Present real-time readings and detect anomalies
Tools: Grafana, Thingsboard, Kibana, Node-RED Dashboard
Methods: Graphs, heat maps, real-time dashboards, and alerts
Interaction and Remote Control
Interaction Types
Human-to-Device (H2D): Mobile apps and voice assistants
Device-to-Device (D2D): Autonomous automation and coordination
Device-to-Cloud (D2C): Sensor data sent for analytics
Cloud-to-Device (C2D): Control commands sent to devices
Remote Control Protocols
HTTP/HTTPS for REST-based commands
MQTT for lightweight publish-subscribe control
CoAP for low-power devices
WebSockets for full-duplex live control
Security Measures
Authentication via login and API tokens
Encryption with TLS/SSL
Access Control Lists (ACLs) and activity logging
IoT Use Cases
Industrial Automation (IIoT)
Components: Sensors, PLCs, industrial fieldbuses
Applications: Predictive maintenance and asset tracking
Commercial Building Automation
Features: Smart lighting, HVAC, smart meters
Protocols: BACnet, KNX, ZigBee, Z-Wave
Raspberry Pi Remote Access
SSH protocol for command-line access
VNC for full desktop GUI access
Tools: AnyDesk, TeamViewer, BalenaCloud
Smart Cities
Architecture: Perception, Network, Edge, Application, User Interface
Smart Traffic Management with adaptive signals
Smart Waste Management with fill-level sensors
Smart Water Supply with leakage detection